

STATISTICS FOR DATA SCIENCE

Complete Training Notes — Easy Explanations, Corrected Concepts, Examples, Formulas and Python Reference

Topics: Descriptive & Inferential Statistics • Population & Sample • Data Types • Levels of Measurement • Central Tendency • Dispersion • Skewness • Probability Distributions • Empirical Rule • Z-score & Standardization

1. What is Statistics?

Statistics is the science of collecting, organizing, summarizing, analyzing, interpreting, and drawing conclusions from data.

In simple words, statistics helps us answer: *What is happening in the data? What patterns exist? How much does the data vary? How certain are we about a conclusion?*

Why Statistics Matters in Data Science

- **Understand patterns:** Find trends, unusual behavior, relationships, and distributions.
- **Summarize large datasets:** Convert many observations into understandable measures.
- **Support decisions:** Use evidence instead of guesses.
- **Prepare for machine learning:** Understand features, variability, outliers, scaling, and uncertainty.
- **Forecast:** Historical patterns may support estimates of future outcomes, while still recognizing uncertainty.

Example: With three years of sales data, statistics can summarize average sales, identify seasonal peaks, measure variability, compare regions, and support forecasting.

2. Descriptive vs Inferential Statistics

2.1 Descriptive Statistics

Descriptive statistics summarizes the data that has actually been observed. It tells us what the available dataset looks like.

- Mean, median and mode
- Minimum, maximum and range
- Variance and standard deviation
- Percentiles and quartiles
- Tables, charts and distributions

Example: In a survey of 100 people, 20 preferred blue cars. The descriptive conclusion is: *20% of the surveyed sample preferred blue cars.*

2.2 Inferential Statistics

Inferential statistics uses a **sample** to estimate, test, or draw conclusions about a larger **population**. Because only part of the population is observed, uncertainty must be measured.

- Confidence intervals
- Hypothesis testing
- Estimation
- Z-tests and t-tests
- Probability-based inference

Correction: Observing 20% in the sample is descriptive. Estimating the population percentage with a confidence interval or margin of error is inferential.

Descriptive	Inferential
Summarizes observed data	Generalizes from sample to population
What happened in this dataset?	What can we infer about the population?
Mean, median, charts, standard deviation	Confidence intervals, tests, estimation

3. Population, Sample and Core Terminology

Term	Easy Meaning	Example
Population	The complete group of interest	All customers of a company
Sample	A subset selected from the population	500 selected customers
Variable	A characteristic measured for each observation	Age, income, color
Observation	One recorded unit or row	One customer's record
Dataset	A collection of observations	A customer table
Experiment / Study	A structured process used to collect or analyze evidence	Testing a treatment
Parameter	A numerical measure describing a population	True population mean
Statistic	A numerical measure calculated from a sample	Sample mean

Important: A parameter describes a population. A statistic describes a sample and is often used to estimate an unknown population parameter.

4. Types of Data

4.1 Numerical / Quantitative Data

Numerical data represents quantities and supports meaningful arithmetic operations.

Discrete Data

Discrete data consists of countable values, usually whole-number counts.

- Number of students
- Number of products purchased
- Number of calls received
- Number of complaints

Continuous Data

Continuous data can take any value within a range, including decimal values.

- Height: 170.2 cm
- Weight: 73.45 kg
- Time: 12.81 seconds

- Distance: 4.728 km

4.2 Categorical / Qualitative Data

Categorical data represents labels, names, classes, or groups.

- Car brand: Audi, BMW, Mercedes
- Department: Sales, HR, IT
- Yes / No
- City or country

5. Levels of Measurement

Scale	Meaning	Order?	True Zero?	Example
Nominal	Categories only	No	No	Color, country, brand
Ordinal	Ordered categories	Yes	Not required	Low/Medium/High
Interval	Equal intervals, no true zero	Yes	No	Temperature in °C or °F
Ratio	Equal intervals with true zero	Yes	Yes	Height, weight, age

Correction: Celsius temperature is an **interval** scale, not a ratio scale. Height and weight are ratio variables because a meaningful zero exists.

6. EDA: Start With the Problem

- **Step 1:** Understand the client or business requirement.
- **Step 2:** Identify whether the task is analysis, regression, classification, forecasting, or another problem.
- **Step 3:** Understand every feature and its data type.
- **Step 4:** Check missing values, duplicates, invalid values, and outliers.
- **Step 5:** Study central tendency, spread, distributions, and relationships.

Practical rule: The problem statement tells you what matters. The data tells you what is possible.

7. Measures of Central Tendency

Central tendency describes a representative or central value of a dataset.

7.1 Mean

The mean is the arithmetic average: **sum of all observations ÷ number of observations**.

Best when numerical data is reasonably symmetric and not dominated by severe outliers.

7.2 Median

The median is the middle value after sorting. For an even number of observations, average the two middle values.

The median is especially useful for skewed data or data with influential outliers.

7.3 Mode

The mode is the most frequently occurring value. It is useful for both repeated numerical values and categorical data.

Measure	Meaning	Outlier Sensitivity	Common Use
Mean	Arithmetic average	High	Roughly symmetric numerical data
Median	Middle ordered value	Low	Skewed data / outliers
Mode	Most frequent value	Not affected in the same way	Categorical or repeated values

```
# Import Python's built-in statistics module
import statistics

# Sample numerical observations
data = [150, 123, 134, 170, 146, 124, 113]

# Calculate the arithmetic average
mean_value = statistics.mean(data)

# Find the middle value after sorting
median_value = statistics.median(data)

print("Mean:", mean_value)
print("Median:", median_value)
```

8. Measures of Dispersion

Dispersion tells us how widely observations are spread.

8.1 Range

Range = Maximum – Minimum

Example: minimum = 113 and maximum = 170, so range = 57.

8.2 Variance

Variance measures the average squared distance of observations from the mean. Larger variance generally means greater spread.

- Find the mean.
- Calculate each observation's deviation from the mean.
- Square each deviation.
- Average the squared deviations.

8.3 Standard Deviation

Standard deviation is the square root of variance and expresses spread in the same units as the original data.

Correction: Small standard deviation = observations are more concentrated around the mean. Large standard deviation = observations are more spread out.

```
# NumPy is widely used for numerical calculations
import numpy as np
```

```
data = np.array([113, 123, 124, 134, 146, 150, 170])

# Maximum minus minimum
data_range = np.max(data) - np.min(data)

# Population variance
variance = np.var(data)

# Population standard deviation
std_dev = np.std(data)

print("Range:", data_range)
print("Variance:", variance)
print("Standard deviation:", std_dev)
```

9. Symmetric and Skewed Distributions

9.1 Symmetric Distribution

A symmetric distribution has a similar shape on both sides of its center. In a perfectly symmetric unimodal distribution, mean, median, and mode coincide.

9.2 Right / Positive Skew

The long tail extends toward larger values. High extreme observations pull the mean to the right. A common pattern is **Mode < Median < Mean**.

Example: income data with a small number of extremely high earners.

9.3 Left / Negative Skew

The long tail extends toward smaller values. Low extreme observations pull the mean to the left. A common pattern is **Mean < Median < Mode**.

Example: an easy exam where most students score highly but a few score very low.

Remember: Skew direction is named after the direction of the long tail.

10. Outliers

An outlier is an observation unusually far from the majority of the data.

- A genuine rare event
- A data-entry error
- A measurement error
- Fraud or anomaly
- A separate population or process

Do not automatically delete outliers. Investigate them first. A valid extreme observation may contain the most important information in the dataset.

11. Probability Distributions

A probability distribution describes how probability is assigned across possible values of a random variable.

11.1 Normal Distribution

A continuous, symmetric, bell-shaped distribution defined by its mean and standard deviation.

11.2 Bernoulli Distribution

Models **one binary trial**: success/failure, yes/no, 1/0, or one head/tail outcome.

11.3 Binomial Distribution

Models the **number of successes in a fixed number of independent Bernoulli trials** with constant success probability.

Example: probability of exactly 6 heads in 10 fair coin tosses.

11.4 Uniform Distribution

Represents equal likelihood across the defined outcomes or range under the model.

- Fair die: each face has probability 1/6.
- A continuous interval can be modeled as uniform if every point in the interval is assumed equally likely.

Caution: Real bus arrival times are not automatically uniform. Uniformity is an assumption that must be justified.

11.5 Poisson Distribution

Models the number of events occurring in a fixed time or space interval under assumptions such as a stable average rate and approximately independent occurrences.

- Calls received per hour
- Defects per square meter
- Arrivals during a fixed period

Distribution	Situation	Example
Normal	Continuous measurements	Height or measurement errors
Bernoulli	One binary trial	One pass/fail outcome
Binomial	Success count in fixed n trials	Heads in 10 tosses
Uniform	Equal likelihood under model	Fair die
Poisson	Event count in fixed interval	Calls per hour

12. Empirical Rule: 68–95–99.7

For an approximately normal distribution:

- About **68%** of observations lie within 1 standard deviation of the mean.
- About **95%** lie within 2 standard deviations.
- About **99.7%** lie within 3 standard deviations.

Observations beyond ± 3 standard deviations are relatively rare under an ideal normal model, but they should not automatically be treated as errors.

13. Standard Normal Distribution and Z-score

Features can use completely different units, such as centimeters, kilograms, dollars, and blood-pressure measurements. For many machine-learning algorithms, scaling features to a comparable numerical scale is useful.

Z-score formula: $z = (x - \mu) / \sigma$

- x : original observation
- μ : mean
- σ : standard deviation
- z : standardized distance from the mean

A standardized feature has mean approximately 0 and standard deviation approximately 1.

Correction: A Z-score is **not itself a probability**. It tells how many standard deviations an observation is above or below the mean. A probability may then be calculated from a specified distribution.

Z-score	Interpretation
0	At the mean
+1	One standard deviation above the mean
-1	One standard deviation below the mean
+2.5	2.5 standard deviations above the mean
-3	Three standard deviations below the mean

Python: StandardScaler

```
# Import the standardization tool
from sklearn.preprocessing import StandardScaler
import pandas as pd

# Features are measured in different units
data = pd.DataFrame({
    "height_cm": [160, 170, 180, 175],
    "weight_kg": [55, 70, 85, 78]
})

# Create the scaler
scaler = StandardScaler()

# Learn each feature's mean and standard deviation,
# then transform the values
scaled_values = scaler.fit_transform(data)

# Convert the result back to a DataFrame
scaled_data = pd.DataFrame(
    scaled_values,
    columns=data.columns
)
```



```
print(scaled_data)
```

Machine-learning best practice: Split the data first. Fit the scaler only on training data, then use the same fitted scaler to transform test data. Otherwise, information from the test set can leak into preprocessing.

```
# Correct scaling workflow
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Split the dataset before learning preprocessing parameters
X_train, X_test = train_test_split(
    data,
    test_size=0.25,
    random_state=42
)

scaler = StandardScaler()

# Learn mean and standard deviation from training data only
X_train_scaled = scaler.fit_transform(X_train)

# Apply the same learned transformation to test data
X_test_scaled = scaler.transform(X_test)
```

14. Complete Data-Science Thinking Flow

Step	Question
1. Problem Statement	What problem am I solving?
2. Understand Data	What does each row and column represent?
3. Data Types	Numerical, categorical, discrete, continuous?
4. Measurement Scale	Nominal, ordinal, interval, or ratio?
5. Data Quality	Missing values, duplicates, invalid records, outliers?
6. Central Tendency	Where is the center?
7. Dispersion	How widely is the data spread?
8. Distribution	Normal, skewed, binary, count-based, etc.?
9. Relationships	How do features relate to each other and the target?
10. Preprocessing	Scaling, encoding, transformation, outlier strategy?
11. Modeling	Which algorithm matches the problem?
12. Evaluation	Does the model generalize to unseen data?

15. Interview-Ready Answers

What is descriptive statistics?

Descriptive statistics summarizes and describes observed data using measures such as mean, median, standard deviation, tables, and visualizations.

What is inferential statistics?

Inferential statistics uses sample data to estimate, test, or draw conclusions about a larger population while accounting for uncertainty.

When should you use median instead of mean?

I prefer the median when numerical data is strongly skewed or contains influential outliers because the median is more robust.

What is standard deviation?

Standard deviation measures how spread out observations are around the mean and is expressed in the same units as the original variable.

What is skewness?

Skewness describes asymmetry. A right skew has a longer right tail, while a left skew has a longer left tail.

What is a Z-score?

A Z-score tells how many standard deviations an observation lies above or below the mean.

Bernoulli vs binomial?

Bernoulli models one binary trial. Binomial models the number of successes across a fixed number of independent Bernoulli trials with constant success probability.

16. Final Revision Cheat Sheet

- **Statistics:** collect, organize, analyze, interpret, and draw conclusions from data.
- **Descriptive:** summarizes observed data.
- **Inferential:** uses a sample to infer about a population.
- **Population:** entire group; **Sample:** subset.
- **Parameter:** population measure; **Statistic:** sample measure.
- **Numerical data:** discrete or continuous.
- **Categorical data:** labels or groups.
- **Measurement scales:** nominal, ordinal, interval, ratio.
- **Central tendency:** mean, median, mode.
- **Dispersion:** range, variance, standard deviation.
- **Skewness:** asymmetry of a distribution.
- **Normal:** symmetric bell-shaped distribution.
- **Bernoulli:** one binary trial.
- **Binomial:** success count across fixed repeated trials.
- **Uniform:** equal likelihood under the model.

- **Poisson:** event counts in a fixed interval.
- **Empirical rule:** approximately 68%, 95%, 99.7% within 1, 2, and 3 SDs for normal data.
- **Z-score:** standardized distance from the mean.
- **StandardScaler:** commonly transforms each feature to mean 0 and standard deviation 1.

Best study method: For every topic, remember five things: What is it? Why is it needed? When is it used? What is one real-world example? What is one limitation or assumption?